

Inter-species competition of surface bacterial flora of pomegranate and their role in spoilage

Indu Pant

Bhabha Atomic Research Centre

Shashidhar Ravindranath (✉ shashi@barc.gov.in)

Bhabha Atomic Research Centre

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Abstract

The surface of fruits is heterogenous in term of its microenvironment hence dictate the kind of microflora that develops during storage. A better understanding of spoilage organisms would lead to better preservation methods. The pomegranate was chosen, since its sturdy and spoils slow at room temperature and is ideal for studying fruit spoilage in-situ. In the current study we isolated organisms from fruit surface and study the spoilage and competition amongst microbial species. Total 17 unique bacterial isolates from pomegranate were identified. The 16s rRNA gene identification placed them in 8 major genera (*Acinetobacter*, *Micrococcus*, *Pantoea*, *Microbacterium*, *Strenotrophomonas*, *Bacillus*, *Staphylococcus* and *Exiguobacterium*). Competition assay among isolate suggested that *Exiguobacterium* is dominant species followed by *Micrococcus*, *Pantoea* and *Bacillus*. The consortium of 3 different combinations (5 bacteria each) of isolated bacteria showed the spoilage phenotype on pomegranate. Except for 3 bacterial isolates, the rest of the isolates produced any one or multiple enzymes associated with the food spoilage (cellulase, amylase, lactase, pectinase and protease). The isolates were checked for the presence of genes associated with antibiotic resistance and 78.9% of the tested micro-organisms were *bla*TEM positive. Aminoglycoside resistance genes were present in 10% of the tested microbes. This study demonstrated interspecies competition amongst spoilage organisms. This understanding of surface flora of fruit would give better insights to preserve fruits.

Introduction

Fruits and vegetables are perishable commodities. The spoilage of fruits and vegetables could be physical, physiological, and microbial. The spoilage also depends on both intrinsic and extrinsic factors. Amongst various fruits, pomegranate is a subtropical fruit crop. India is one of the largest producers and exporters of pomegranate (APEDA). There are few studies on the kernels of pomegranate. Krisch et al. have shown that microbes and yeast are responsible for spoilage (Krisch et al. 2016). However, there are very few studies on the spoilage of the whole pomegranate fruit. A better understanding of the fruit spoilage process is essential to ensure good shelf life. The surface bacterial flora of fruit can cause physiological changes in the fruits. The surface associated bacteria are often the reason for spoilage (Gram et al. 2002). The microbial interaction could be synergistic or antagonistic. Food spoilage is often attributed to the dominant microbial flora present on the surface of the fruit.

During the revolution of food processing, many research papers were published on spoilage organisms. The advent of modern food processing and cold supply chains resulted in better preservation of fruits and vegetables. Hence, the focus of food microbiology was more on pathogenic bacteria rather than food spoilage organisms. Presently, sufficient studies focus on metagenomics. However, the physiology and spoilage capabilities of these organisms are unknown. It is essential to study the spoilage with culturable bacteria, the interaction between bacteria and the physiology of the microbes on the fruit. Therefore, in the current study, we have evaluated culturable microbial flora present on the surface of the pomegranate and the role of these microbes in the fruit's spoilage and their importance in the spread of antibiotic resistance genes. As per our knowledge, this is the first report on pomegranate fruit spoilage by microbes.

Material and Methods

Sample Collection and isolation of microbial flora

Fresh pomegranates (20 in number) were purchased from the local fruit market in Vashi, Mumbai. Microbial flora associated with the surface was isolated using the swabbing method (Leff et al. 2013). The cotton swab bud was applied on the pomegranate surface and then rubbed on the plate count agar surface. These plates were incubated at 37°C for 24–48 hours. The different colonies that appeared on the plate were purified and stored as glycerol stock.

Estimation of bacteria, yeast and mold on the pomegranate surface

The estimation of bacterial, yeast and mold was done on plate count agar and potato dextrose agar respectively by using spread plate technique. The plates were incubated at 37°C and 26°C respectively and counting of bacteria, yeast and mold on the plates was done after 24 hours and 48 hours respectively.

Identification using 16srRNA sequencing

The pure cultures were identified by the 16S rRNA method, as described previously (Nossa et al. 2015). The primers used were 347F and 803R. The sequence of 347F is 5'-GGAGGCAGCAGTRGGAAT-3' and 803R is 5'-CTACCRGGGTATCTAATCC-3'. PCR programme used for amplification was (95°C for 3 mins, 95°C for 30 sec, 55°C for 1 min, 72°C for 1 min and final extension at 72°C for 10 mins). PCR product was run on 1% agarose gel. Amplified product was of 450 bp length. The amplified product was cut from the gel and Qiagen Gel extraction kit was used for purification of DNA fragments. The purified DNA fragments of 100-200ng/ul concentration were sent for sequencing.

Identification of isolates for production of enzymes

The microbes isolated were tested for the production of cellulase, protease, amylase, lactase and pectinase activity. In brief, the overnight microbial culture of isolated bacteria was taken and 10 microliters of each bacterial suspension was spot plated on carboxymethyl cellulose, starch agar, pectin agar, lactose agar and casein agar plates (Gebreyohannes et al. 2015; Islam et al. 2018; Leff et al. 2013) and kept for incubation at 37°C and were checked for the zone of hydrolysis after 24 hours.

Evaluation of competitive fitness and spoilage potential among isolates

The competitive fitness and spoilage-causing potential among isolates were also studied. For competitive fitness studies, different combinations of bacterial isolates were inoculated in equal amounts (10^7 CFU/ml) into Luria broth and incubated at 37°C. The competitive fitness was measured by counting CFU/ml

after different time intervals (0th day and 10th day). For spoilage studies, a distinct combination of bacteria (Table 3) was spotted on the fruit surface and observed for black spot appearance over a period of different time interval.

Determination of antibiotic resistance profile of isolates

The Kirby-Bauer disk diffusion method was used to evaluate resistance profile of isolates (Biemer 1973). Detection of antimicrobial resistance genes in the pure cultures was also carried out as described previously (Naik et al. 2018) for the 6-antibiotic resistance-associated genes (*bla*_{TEM}, *tetA*, *gyrA*, *floR*, *cat* and *aadA*).

Statistical Analysis

All the experiments were performed in triplicates and mean are given wherever needed.

Results and discussion

The choice of the fruit pomegranate for spoilage studies was based on two reasons. There are no studies on pomegranate fruits. The second reason is, it is a sturdy fruit and is available throughout the year in Maharashtra, India. The pomegranate fruits can be kept for up to 12 days at room temperature. Other fruits like mango, banana or papaya will spoil soon at room temperature and is difficult to detect spoilage because of microbes or the fruit physiology. Therefore, by using pomegranate fruit, it's easier to evaluate the spoilage potential of individual bacteria or consortia and spoilage can also be proved by Koch's postulates.

This study shows that the pomegranate surface is a habitat for a wide range of bacteria. The microbial load on the pomegranate surface was 10^3 cfu/g and the total yeast and mould count was 10^2 cfu/g. The microbial spectrum could vary with the source of fruit orchard, handling ways and season. Therefore, 20 fruits bought from different sources were used to collect the surface microflora. From each fruit, a swab was spread on the plate. Based on the colony characteristics (shape, colour and serration) total 17 unique colonies were isolated (Table 1). Most of the colonies (> 70%) were white, opaque, mucoid and round. Each of these bacterial colonies was identified by 16S rRNA sequence. The detailed identification is given in Table 1. Diverse bacterial genera were identified from the pomegranate surface as can be seen from phylogenetic tree (Fig. 1a). The phylogenetic tree was generated using Molecular Evolutionary Genetics Analysis version 11 (Tamura, Stecher, and Kumar 2021). These belong to *Firmicutes*, *Enterobacteriaceae*, *Micrococcaceae*, *Microbacteriaceae*, *Xanthomonadaceae* and *Gammaproteobacteria* family (Fig. 1b). Major genera associated with the surface were *Acinetobacter*, *Micrococcus*, *Pantoea*, *Microbacterium*, *Stenotrophomonas*, *Bacillus*, *Staphylococcus* and *Exiguobacterium*. The sequences were submitted to Genbank with accession number MN173078-MN173096. The maximum representation is from the *Bacillus* genus (9 species). There was no significant difference in bacterial diversity between individual fruits studied. However, we cannot rule out the possibility of variation in the bacterial diversity of pomegranate fruits sourced from different geological locations and time periods. But, the bacterial species isolated and reported here are most likely predominant species. Since numerous reports have reported a similar species composition on the other fruit surface (García et al. 2013 ; Leff et al. 2013; Mamphogoro et al. 2020). The metagenomic study across different time points and geographical locations may give much better insight into the microbial diversity of pomegranate.

Table 1

Microbes isolated from pomegranate surface and their potential to produce different enzymes and cause spoilage. "Positive" mean production of particular enzyme by the organism.

Putative microbes	Cellulase	Amylase	Protease	Lactase	Pectinase	Spoilage Organism
<i>Acinetobacter schindleri</i>				Positive		Yes
<i>Bacillus megaterium</i>	Positive	Positive			Positive	Yes
<i>Bacillus endophyticus</i>		Positive	Positive			Yes
<i>Micrococcus luteus</i>					Positive	No
<i>Bacillus aryabhattai</i>	Positive	Positive	Positive			Yes
<i>Bacillus spp.</i>		Positive	Positive			Yes
<i>Pantoea dispersa</i>						Yes
<i>Bacillus anthracis</i>	Positive					Yes
<i>Staphylococcus saprophyticus</i>						No
<i>Bacillus anthracis</i>	Positive		Positive			Yes
<i>Exiguobacterium enclense</i>		Positive				Yes
<i>Enterobacter cloacae</i>						No
<i>Acinetobacter calcoaceticus</i>	Positive					Yes
<i>Bacillus cereus</i>	Positive	Positive			Positive	Yes
<i>Bacillus subtilis</i>	Positive	Positive				Yes
<i>Stenotrophomonas maltophilia</i>			Positive			Yes
<i>Microbacterium barkeri</i>	Positive					Yes

However, the aim of the current study is to isolate culturable bacteria and study spoilage physiology. Out of 17 species identified, 10 are known for food spoilage (Table 1), which gives us a clue that this normally present flora on the fruit surface can lead to spoilage of the fruit under certain conditions. Food-borne pathogens associated with genera were low; only *Staphylococcus* and *Enterobacter* were detected. It was shown that differences in relative abundance in microbial composition can also be attributed to changes in metabolites, physical interaction and symbiotic interactions between the host surface and other surface inhabitants (García et al. 2013). The bacteria on fruit or any other surface compete for space and nutrients. The complex dynamics of these bacterial communities decide the fruit spoilage. Since many of the bacteria cannot be cultured, it is difficult to understand the physiology of these bacterial interactions. However, in the current study, we could select four cultures with distinct colony morphology for interaction studies. In each set, two bacteria were co-inoculated in the Luria broth (LB) media and survival was observed by the plating method. Since we can distinguish the colonies, their survival could be monitored. A total of 12 sets of this co-culture were set up (Table 2) and observed for at least 10 days at room temperature. *Exiguobacterium enclense* out-competed all four tested species. *Micrococcus luteus* inhibited *Pantoea dispersa* and *Bacillus megaterium*. *Pantoea dispersa* survived better than *Bacillus megaterium*. *Exiguobacterium enclense* and *Bacillus megaterium* both are reported as the plant promoting bacteria in rhizosphere. However, there are no reports on the implications of these bacteria in food spoilage. *Exiguobacterium enclense* produces amylase and *Bacillus megaterium* produces cellulase, amylase and protease. Hence, both bacteria may contribute to fruit spoilage. Further, the four bacteria (*Exiguobacterium enclense*, *Pantoea dispersa*, *Bacillus megaterium* and *Micrococcus luteus*) have been reported to produce different bioactive compounds (Matar et al. 2009; Shanthakumar et al. 2015; Umadevi et al. 2013; Walterson et al. 2014). These bacteria may inhibit the quorum sensing and hence survive better than the other bacteria. These cultures may interact in different modes on the fruit surface. Since fruit surfaces may promote or inhibit a certain set of bacterial populations (Leff et al. 2013). However, these co-culture studies were not done on the fruit. Since the uniform distribution of competing cultures and retrieval of these cultures at a regular time point is difficult to reproduce. But the current co-culture studies give insights into inter-species interaction. However, multiple cultures with green fluorescent markers and continuous monitoring with time-lapse microscopy may shed more insights into the true nature of these bacterial interactions on the fruit surface.

Table 2

Competition among different microbes isolated from pomegranate surface and their survival in Luria broth in competition with other microbes for 10 days. The competition experiment suggest *Exiguobacterium* is most dominant species followed by *Micrococcus*, *Pantoea* and *Bacillus*. The order of dominance is *Exiguobacterium* > *Micrococcus* > *Pantoea* > *Bacillus*

Competition between Organisms	Organism survived in competition after 10 days
<i>Exiguobacterium</i> + <i>Micrococcus luteus</i>	<i>Exiguobacterium enclense</i>
<i>Exiguobacterium</i> + <i>Pantoea dispersa</i>	<i>Exiguobacterium enclense</i>
<i>Exiguobacterium</i> + <i>Bacillus megaterium</i>	<i>Exiguobacterium enclense</i>
<i>Micrococcus luteus</i> + <i>Pantoea dispersa</i>	<i>Micrococcus luteus</i>
<i>Micrococcus luteus</i> + <i>Bacillus megaterium</i>	<i>Micrococcus luteus</i>
<i>Pantoea dispersa</i> + <i>Bacillus megaterium</i>	<i>Pantoea dispersa</i>

In the literature, many bacterial species are reported associated with food spoilage. Though seldom studies tried to prove Koch's postulates and prove the causative organism. We attempted inoculating all 17 species, independently on the fruit surface. However, there was no significant spoilage observed after 10 days. Hence, we made 3 different combinations of 5 bacteria each (Table 3). These bacterial combinations were inoculated on the alcohol-sterilized pomegranate surface. All the 3 combinations were found to spoil the fruit within 10 days. The black spots appeared on the locations where bacterial cocktail was applied. We could also isolate inoculated microbes from the black spot region which proves that spoilage is caused by inoculated microbes and is not the result of any other microbial contamination (Fig. 2). The major reason of the spoilage could be the presence of *Bacillus* genus as most of the isolates belong to this genus, and the genus is already known for spoilage of fruits and vegetables (Tewari et al. 2015). All the *Bacillus* species isolated in the current study are producers of one or the other food spoilage enzymes (Table 1). But this is also possible that the combination of a set of bacteria may cause more damage than the other set. Often, food spoilage organisms produce various extracellular enzymes for the degradation of the food. In the current study, we found that 35% of the isolated bacteria produced cellulase, 41% produced amylase, 5% produced lactase, 17% produced pectinase and 26% were positive for the production of protease enzymes (Fig. 3). We found that *Bacillus* spp. were producing a maximum amount of amylase, cellulase and pectinase and this could be one reason for the predominant presence of the *Bacillus* genus on the fruit surface.

Table 3

Black spot produced by different combination of microbes isolated from surface of pomegranate. +ve indicate presence of black spot which means infection occurred.

Isolate Combination	Black Spot Detection
<i>Acinetobacter pittii</i> + <i>Micrococcus luteus</i> + <i>Bacillus megaterium</i> + <i>Pantoea</i> + <i>Staphylococcus saprophyticus</i>	+ve
<i>Exiguobacterium enclense</i> + <i>Micrococcus luteus</i> + <i>Bacillus megaterium</i> + <i>Microbacterium barkeri</i> + <i>Stenotrophomonas maltophilia</i>	+ve
<i>Pantoea dispersa</i> + <i>Microbacterium barkeri</i> + <i>Bacillus megaterium</i> + <i>Exiguobacterium enclense</i> + <i>Stenotrophomonas maltophilia</i>	+ve

Hence, any bacterial species which has an inhibitory effect on the *Bacillus* species could be a good bio-control agent.

The food spoilage organism and antibiotic resistance are not related. However, there are many reports on an increase in the antibiotic resistance gene pool in the environmental microbiome. In 2021, Larsson et al., has indicated the importance of the environment in the evolution of resistance (Larsson et al. 2022). The fruits are often treated with pesticides and this also may lead to the selection of an antibiotic-resistant population (Jeadran et al. 2020). Another major concern is the horizontal gene transfer from these resistant isolates to the sensitive bacteria (Von et al. 2016; Larsson et al. 2022). Hence, bacterial isolates from the pomegranate were evaluated for different antibiotic resistance genes and antibiotic resistance profile is given in Table 4. A few of them were also tested positive for antibiotic resistance genes. These bacteria were found to contain *blaTEM* and *aadA* genes (Ramirez et al. 2010). *blaTEM* gene encodes resistance against β lactam antibiotics, e.g. Penicillin, cephalosporin etc. (Bajaj et al. 2015; Lachmayr et al. 2009). Among the tested microorganisms, 78.9% were *blaTEM* positive whereas aminoglycoside resistance genes were present in 10% of the tested microbes.

Table 4

Antibiotic resistance profile of microbes isolated from pomegranate surface. The zone of inhibition was measured in “cm”. The zone of inhibition indicate the sensitive to the antibiotics where “R” represents resistance of microbe to that particular antibiotic.

Organism	Ampicillin (cm)	Chloramphenicol (cm)	Vancomycin (cm)	Methicillin (cm)	Rifampicin (cm)	Ofloxacin (cm)	Streptomycin (cm)	Ticarcillin (cm)	Cephalothin (cm)
<i>Acinetobacter schindleri</i>	R	2	0.8	R	1.5	2.1	R	3	R
<i>Bacillus megaterium</i>	1.7	2.6	2	R	1.9	2.6	4	R	3
<i>Bacillus endophyticus</i>	1.8	3	1.6	R	1.8	2.5	1	R	2
<i>Micrococcus luteus</i>	1	1	2	R	1	1	2	R	3
<i>Bacillus aryabhatai</i>	1	3	1.7	R	2.1	2.5	2.1	R	4
<i>Bacillus spp.</i>	1	1.8	1.9	R	1	1.7	2	R	3
<i>Pantoea dispersa</i>	R	2	3.5	1	2	1	2	R	2
<i>Bacillus zanthoxyli</i>	1.8	1.8	2.5	R	2	1.6	1	R	2
<i>Staphylococcus saprophyticus</i>	2.1	3	2.2	R	1.8	1.6	1	R	2
<i>Bacillus acanthi</i>	1	0.9	3	R	2	2.9	2	R	2
<i>Exiguobacterium enclense</i>	3	2.5	1.4	R	2	2	2	R	2.8
<i>Enterobacter cloacae</i>	R	R	R	R	R	R	R	R	R
<i>Acinetobacter calcoaceticus</i>	0.8	3	2	2	1.1	R	R	R	R
<i>Bacillus pseudomycoides</i>	2	1	1	1	1	.6	2	1.1	1
<i>Bacillus subtilis</i>	2	2.1	1.9	0.4	1.1	1	2.3	2	1
<i>Strenotrophomas maltophilia</i>	1	1.0	1.6	2	1.6	2	1.6	.6	.9
<i>Microbacterium barkeri</i>	1	0.6	0.2	1.1	1.2	1.9	2	2.1	2.2

The current study couldn't address a few questions. Is the inter-species competition on the fruit similar to that observed in the growth medium? The number of permutations and combinations of the 17 culturable bacteria is huge. Therefore, all the combinations were not tested for fruit spoilage.

This study shed light on the ecology of the culturable microbiome of the pomegranate fruit, which may be responsible for the spoilage of pomegranate fruit. We attempted to assess competitive fitness among different spoilage-associated isolates and it was found that *Exiguobacterium* is the dominant species followed by *Micrococcus*, *Pantoea* and *Bacillus*. The reason for *Exiguobacterium* being a dominant species in a competitive environment might lie in its ability to survive in varying environments ranging from different pH, and temperature to high salt and metal stress. These isolates can also produce hydrolytic enzymes. The study of *blaTEM* gene containing environmental reservoirs of bacteria on the fruit surface will be helpful in devising strategies to protect the public from the menace of clinical risks linked with antimicrobial-resistant bacteria.

Abbreviations

blaTEM -β-Lactamase

aadA1 - aminoglycoside (3") (9) adenylyltransferase

Declarations

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Author's contribution

Dr. Ravindranath Shashidhar - Conceptualization of the idea, writing, review and editing.

Miss. Indu Pant- Working, writing, review, editing.

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Conflict of interest

The authors declare no conflict of interest.

Ethical Approval

This study did not contain any data associated with human and animal studies.

Consent to participate

Not Applicable

Consent for publication

Not applicable

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- ## Figures

Figure 1

(a) Phylogenetic tree of bacteria isolated from pomegranate surface (b) Diversity of bacterial families isolated from pomegranate surface

Figure 2

Progression of infection on pomegranate surface with inoculated isolates

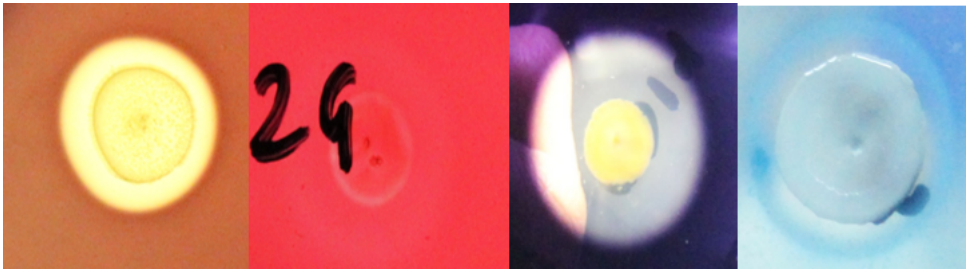


Figure 3

Representative pictures of isolates producing different enzymes